SEC P vs NP log 2026-05-21

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 12:07 UTC

SEC P vs NP — daily log 2026-05-21

42 cycles recorded

Noncommutative Algebraic Growth and Tseitin Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Algebra × Resolution Width of Tseitin Formulas
- **Recorded**: 2026-05-21 00:30 UTC
- **Entry ID**: `e87da145bc32`

Statement

For a random 3-CNF formula Φ with n variables, let A_Φ be the noncommutative algebra generated by variables x_i and relations x_i x_j = x_j x_i if clauses (i,j) are satisfied. The growth rate of dim(A_Φ_k) is Ω(2^{k/2}) for all k ≤ log n iff Φ's Tseitin formula has resolution width ≥ log n.

Rationale

Noncommutative algebras encode clause interactions via generators, while resolution width measures proof complexity. The conjecture links algebraic growth rates to proof complexity via commutativity constraints, revealing structural trade-offs between algebraic dependencies and proof efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `27.65s`


```
0.0, 'instances_tested': 10, 'conjecture_holds': False, 'counterexample': 'growth_rate_mismatch'}
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
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TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
```

```
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
TRIAL: {'metric_name': 'growth_rate', 'metric_value': 0.0, 'instances_tested': 10, 'conjecture_holds
RESULT: FALSIFIED counterexample="growth_rate_mismatch" first_failing_seed=0
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: The conjecture was falsified by a counterexample showing growth_rate_mismatch in all tested instances. | next: Analyze the specific counterexample to determine how the growth rate deviation correlates with Tseitin resolution width properties

---

Noncommutative Rank Gap in Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Algebra × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 00:52 UTC
- **Entry ID**: `f245272528b8`

Statement

For a read-twice BP P over n variables, the noncommutative rank of its transition matrix M satisfies rank_nc(M) ≥ Ω(n), whereas for read-once BPs, rank_nc(M) = O(log n).

Rationale

Noncommutative rank captures algebraic dependencies in matrix factorizations, which may reflect the repeated variable access in read-twice BPs. This could expose structural asymmetries in their computational power compared to read-once BPs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 105, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 78, in run_trial
    rank = rank_nc(M, p)
           ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 38, in rank_nc
    gaussian_elimination(A, p)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ae8bb1cc.py", line 23, in gaussian_elim
    if abs(A[j][i]) > abs(A[max_row][i]):
       ^^^^^^^^^^^^
TypeError: bad operand type for abs(): 'list'
```

Judge reasoning

Test crashed due to TypeError in Gaussian elimination, preventing result collection. Support condition cannot be evaluated without successful trials. | next: Debug Gaussian elimination implementation to handle matrix entries correctly (ensure numerical values instead of lists)

Non-Abelian Harmonic Analysis and SOS Refutation Degree for CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-Abelian Harmonic Analysis × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-21 01:05 UTC
- **Entry ID**: `93e2fe31dd48`

Statement

For a CSP instance over a non-abelian group G, the SOS refutation degree is bounded by the dual norm of the Fourier coefficients of the CSP's constraint function. Specifically, for every CSP instance φ over G, refutation_degree(φ) ≤ ||F||_∞, where F is the Fourier transform of φ's constraint function.

Rationale

Non-abelian harmonic analysis decomposes CSP constraints into group characters, revealing structural symmetries that may influence the SOS hierarchy's ability to certify unsatisfiability. The dual norm of Fourier coefficients quantifies the 'spread' of constraints across irreducible representations, which could correlate with the minimal SOS degree required for refutation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6c3843cd.py", line 50, in run_trial
    F = [1] * (2 ** n)  # Placeholder Fourier coefficients
        ~~~~^~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of support fraction or counterexamples. | next: Optimize memory usage in test code for large n, then re-run experiments

---

Non-Abelian Fourier Coefficient Gap Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Non-Abelian Harmonic Analysis × Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 01:53 UTC
- **Entry ID**: `324ca13de3e5`

Statement

For any 3-CNF formula φ with n variables, the minimal non-Abelian Fourier coefficient |F[π]| over all irreducible representations π of the symmetric group S_n satisfies |F[π]| ≥ Ω(1/√n) if and only if the disjointness communication complexity of φ is Ω(log n).

Rationale

Non-Abelian Fourier analysis captures group-theoretic symmetries in communication matrices. Disjointness requires coordination across independent sets, which may manifest as nontrivial Fourier coefficients in symmetric group representations, revealing structural constraints on protocol efficiency.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 65, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6de39bfc.py", line 45, in run_trial
    clause = [random.randint(0, n-1) for _ in range(3)]
                                ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable ‘n’ before producing results | next: Fix the NameError in test code by defining ‘n’ before using it in clause generation

---

ε-Approximation Complexity Lower Bound for k-CLIQUE via Monotone DNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Approximation Theory of Boolean Functions × Monotone DNF Formulas for k-CLIQUE
- **Recorded**: 2026-05-21 02:10 UTC
- **Entry ID**: `5c00c6af4f62`

Statement

For any ε ∈ (0, 1/2), the ε-approximation complexity of the k-CLIQUE function is Ω(n) when k ≥ log n, and this complexity is submodular under conjunction of functions.

Rationale

Approximation theory provides tools to analyze how well boolean functions can be approximated by simpler functions, which could reveal structural limitations in monotone DNF representations. The submodularity under conjunction suggests that combining functions increases the required complexity, aligning with known lower bounds for k-CLIQUE.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 49, in run_trial
    clique = generate_k_clique(n, k)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in generate_k_cl
    if all(node not in edge for edge in clique):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e72ffab8.py", line 25, in <genexpr>
    if all(node not in edge for edge in clique):
           ^^^^^^^^^^^^^^^^
TypeError: argument of type 'int' is not iterable
```

Judge reasoning

Test crashed with TypeError due to non-iterable integer in membership check, preventing data collection | next: Fix the generate_k_clique function to ensure edges are iterable structures (e.g., tuples) before membership testing

---

Matroid Rank and ABP Size for Polynomial Representations

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Algebraic Branching Program Size
- **Recorded**: 2026-05-21 02:20 UTC
- **Entry ID**: `d2edbef09287`

Statement

The matroid rank of the coefficient matrix of a polynomial represented by an ABP of size s is $\Theta(\log s)$.

Rationale

Matroid rank captures the linear independence structure of the polynomial's coefficients, which may reflect the minimal complexity of representing the polynomial via an ABP. This bridge could expose hidden algebraic dependencies in the ABP's structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2b11b003.py", line 51, in run_trial
    A = [[coefficients[i] * j**k for k in range(n + 1)] for i in range(n + 1)]
                            ^
NameError: name 'j' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'j', preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_2b11b003.py by defining variable 'j' in the list comprehension

---

Sheaf Cohomology Euler Characteristic and Tseitin Resolution Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sheaf Cohomology × Resolution Proof Size for Tseitin Formulas
- **Recorded**: 2026-05-21 02:31 UTC
- **Entry ID**: `62b0c1d5061d`

Statement

For a Tseitin formula derived from a graph G with n vertices, the absolute value of the Euler characteristic of the sheaf cohomology groups H^i(G) is inversely proportional to the resolution proof size, i.e., |χ(H^i(G))| = Θ(1 / log n).

Rationale

Sheaf cohomology captures the topological structure of the graph, which may influence the complexity of the resolution proof. The Euler characteristic provides a quantitative measure of this topological invariant, potentially linking it to the proof size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ff24d2c.py", line 43, in run_trial
    clauses.append([-literals[i], -literals[j]])
                    ^^^^^^^^^^^^
TypeError: bad operand type for unary -: 'str'
```

Judge reasoning

Test crashed due to type error in clause generation, preventing data collection | next: Fix the TypeError by ensuring literals are integers before applying unary negation

---

Fourier Norm Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Harmonic Analysis on Locally Compact Groups × Randomized Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 03:12 UTC
- **Entry ID**: `af4613884bf0`

Statement

For any n×n communication matrix M of the Disjointness problem, the L^2 norm of its Fourier transform over the additive group Z_n^2 is Ω(n).

Rationale

Harmonic analysis on finite abelian groups captures the spectral structure of communication matrices, with Fourier norms reflecting the 'spread' of information across frequencies. The Disjointness matrix's inherent combinatorial structure may enforce nontrivial Fourier norms, linking algebraic invariants to communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 54, in run_trial
    F = fft_2d(M)
        ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 43, in fft_2d
    M_fft[i] = fft([M[i][j] for j in range(n)])
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 34, in fft
    even = fft([a[k] for k in range(0, n, 2)])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 2 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3a7c6574.py", line 36, in fft
    T = [cmath.exp(-2j * math.pi * k / n) * odd[k] for k in range(n // 2)]
         ^^^^^
NameError: name 'cmath' is not defined. Did you mean: 'math'? Or did you forget to import 'cmath'?
```

Judge reasoning

Test crashed due to missing cmath import, preventing evaluation of Fourier norm bounds. | next: Fix the test by importing cmath and re-running with proper error handling

---

Laplacian Eigenvalue and Resolution Length of Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Graph Theory × Resolution Length of Tseitin Formulas
- **Recorded**: 2026-05-21 03:30 UTC
- **Entry ID**: `9e9baa385874`

Statement

For any graph G, the resolution length of the Tseitin formula on G is at least 2^{Ω(1/(1 - λ_2))}, where λ_2 is the second smallest eigenvalue of the Laplacian matrix of G. For graphs where 1 - λ_2 ≥ c (constant c > 0), the resolution length is exponential in 1/(1 - λ_2).

Rationale

The second eigenvalue of the Laplacian (λ_2) captures the expansion properties of the graph. A smaller λ_2 indicates a better expander, leading to longer resolution proofs as the expansion increases, requiring more steps to refute the Tseitin formula.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 57, in run_trial
    λ2 = min(laplacian_eigenvalues(G)[1:])
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 41, in laplacian_eig
    L = gaussian_elimination(L)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7ed580cd.py", line 28, in gaussian_elim
    A[k][j] += factor * A[i][j]
    ~~~~^^^
```

```
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError in Gaussian elimination, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the Gaussian elimination implementation to handle matrix indices properly and re-run tests with multiple seeds

---

Moment Matrix Spectral Gap and SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 03:38 UTC
- **Entry ID**: `797c606cf901`

Statement

For any n-variable Max-CUT instance, the smallest eigenvalue of its degree-d moment matrix satisfies $\lambda\_min \geq \Theta(1/\sqrt{n})$ iff $d \geq \Omega(\log n)$. The spectral gap $\gamma = \lambda\_max - \lambda\_min$ scales as $\gamma = \Theta(1/n)$ for $d = \Omega(\log n)$.

Rationale

Random matrix theory provides tools to analyze the spectral properties of moment matrices, which are central to SOS hierarchies. The spectral gap directly influences the convergence rate of SOS approximations, linking algebraic structure to algorithmic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 180, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 153, in run_trial
    moment_matrix = construct_moment_matrix(instance, d)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 39, in construct_mom
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1d82cae.py", line 39, in <genexpr>
    m[i][j] += sum(lit ** abs(j - k) for lit in clause)
                                  ^
NameError: name 'k' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'k' in the moment matrix construction, preventing data collection. | next: Debug the NameError in the genexpr by checking variable scoping for 'k' in the clause summation

---

Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 03:52 UTC
- **Entry ID**: `b6ef2369c2bb`

Statement

For a read-twice BP P over n-bit inputs, let μ_P be the non-commutative distribution of its transition matrices. Define ρ(P) = |R(μ_P)(0)|, where R(μ_P) is the R-transform of μ_P. Then ρ(P) ≥ Ω(log n) for all P computing IP_2, but ρ(P) = O(1) for all read-once BPs computing IP_2.

Rationale

Free entropy measures the 'non-commutative complexity' of random variables, which may capture the inherent structure of read-twice BPs that read-once BPs lack. The R-transform's zeroth coefficient encodes global scaling behavior, potentially distinguishing these models via their algebraic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `8.28s`

```
e': Fraction(1, 1), 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'ρ(P) < 3.5'
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
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TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho
TRIAL: {'metric_name': 'ρ(P)', 'metric_value': Fraction(1, 1), 'instances_tested': 30, 'conjecture_ho

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a9497589.py", line 76, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                  ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing reliable data, but stdout suggests counterexample ρ(P) < 3.5. The crash invalidates the counterexample's validity. | next: Re-run test with increased instance count and seed diversity to verify conjecture status

---

Matroid Rank Gap in Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone DNF Formulas for k-CLIQUE
- **Recorded**: 2026-05-21 04:29 UTC
- **Entry ID**: `202b457c781c`

Statement

For a k-CLIQUE indicator function expressed as a monotone DNF with m clauses, the rank of the matroid formed by its clause vectors is Ω(n) if m ≤ n^{O(1)}, and O(log n) if m ≥ $n^{\{\Omega(k}$\{1/4\})\}.

Rationale

Matroid rank captures combinatorial dependencies in clause interactions, which may expose structural barriers to succinct DNF representations. The conjecture ties this invariant to known lower bounds for k-CLIQUE, suggesting rank gaps could amplify hardness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 83, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_115ac035.py", line 56, in run_trial
    matrix = [[int(abs(lit) == var) for var in range(1, n + 1)] for clause in dnf]
                       ^^^
NameError: name 'lit' is not defined. Did you mean: 'List'?
```

Judge reasoning

Test crashed due to NameError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Fix the NameError in test_115ac035.py by defining 'lit' variable before usage

---

Algebraic Shifting Facet Count Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic shifting × Resolution length of Tseitin formulas
- **Recorded**: 2026-05-21 04:56 UTC
- **Entry ID**: `312ea3700c09`

Statement

For a Tseitin formula on a graph G, the resolution length is ≥ 2^{Ω(ν(G))}, where ν(G) is the number of facets in the algebraic shifted complex of G. This holds for all graphs with expansion parameter λ(G) ≥ 1/√n.

Rationale

Algebraic shifting captures combinatorial structure of graphs via simplicial complexes, while resolution length measures proof complexity. The conjecture links structural properties of shifted complexes to proof complexity, leveraging the interplay between algebraic topology and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 105, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in run_trial
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_91e2d45b.py", line 80, in <genexpr>
    expansion_param = sum(len(list(graph.neighbors(v))) for v in range(n)) / (n * (n - 1))
                                   ^^^^^^^^^^^^^^^
AttributeError: 'set' object has no attribute 'neighbors'
```

Judge reasoning

Test crashed due to AttributeError in graph.neighbors() call on set objects, preventing data collection | next: Fix graph representation to use a structure with neighbors() method (e.g., NetworkX Graph) and re-run experiments

---

Symmetric Group Orbit Count Distinguishes Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-21 05:31 UTC
- **Entry ID**: `376a7658c1fe`

Statement

For a read-twice BP P, the number of orbits of its associated polynomial representation under the symmetric group S_n action is Θ(log size(P)) for read-once BPs but Ω(n) for read-twice BPs. The orbit count is strictly greater for read-twice BPs when size(P) ≥ 2^n/2.

Rationale

Invariant theory provides a framework to measure symmetry in polynomial representations, which could expose structural differences between read-once and read-twice BPs. The symmetric group action on variables creates a natural invariant that may scale differently with BP complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `245.34s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Polynomial Threshold Function Degree Bounds Karchmer-Wigderson Protocol Cost for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polynomial Threshold Functions × Karchmer-Wigderson Protocol Cost
- **Recorded**: 2026-05-21 05:41 UTC
- **Entry ID**: `8a180d66cac2`

Statement

For any CNF formula F on n variables computing PARITY, the minimal degree of a polynomial threshold function representing F is $\Omega(\log n)$.

Rationale

Polynomial threshold functions capture the structure of Boolean functions, and their degree may reflect the complexity of communication protocols needed to compute them. The Karchmer-Wigderson protocol cost measures the communication complexity of computing a function, linking algebraic representation to interactive proof strategies.

Novelty

- Judge: `ADJACENT_OK` over 7 arXiv hits

Top hits consulted: - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1504.01064v2] The degree of the Alexander polynomial is an upper bound for the topological slice genus - [1806.07358v1] No Threshold graphs are cospectral - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer

Empirical Test

- exit code: `0`, elapsed: `0.06s`

```
: 1, "conjecture_holds": True, "counterexample": ""}
TRIAL: {"seed": 503, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.700825
TRIAL: {"seed": 547, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.529875
TRIAL: {"seed": 593, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.341542
TRIAL: {"seed": 631, "metric_name": "polynomial_threshold_function_degree", "metric_value": 2.728451
TRIAL: {"seed": 677, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.381222
TRIAL: {"seed": 727, "metric_name": "polynomial_threshold_function_degree", "metric_value": 7.854839
TRIAL: {"seed": 773, "metric_name": "polynomial_threshold_function_degree", "metric_value": 3.234422
TRIAL: {"seed": 821, "metric_name": "polynomial_threshold_function_degree", "metric_value": 8.674023
TRIAL: {"seed": 877, "metric_name": "polynomial_threshold_function_degree", "metric_value": 1.723483
TRIAL: {"seed": 929, "metric_name": "polynomial_threshold_function_degree", "metric_value": 6.568578
RESULT: SUPPORTED mean=4.618178297706176 std=2.272930228914458 support_fraction=0.9
```

Judge reasoning

parse_fail | next: NONE

Fourier Coefficient Sum Bounds Monotone DNF Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis on Boolean Functions × Monotone DNF Size for k-CLIQUE
- **Recorded**: 2026-05-21 06:02 UTC
- **Entry ID**: `fd209c52f373`

Statement

For any monotone DNF formula on n variables, the sum of absolute values of its Fourier coefficients is O(log n). For the k-CLIQUE indicator function, this sum is $\Omega(n)$ for $k \geq 3$.

Rationale

Fourier coefficients capture the influence of variables in DNF formulas. Monotone k-CLIQUE requires exponentially many terms, leading to large coefficient sums, while polynomial-size DNFs have logarithmic sums due to sparsity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`

```
', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
```

```
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
TRIAL: {'metric_name': 'Fourier coefficient sum', 'metric_value': 1, 'instances_tested': 1, 'conjectu
RESULT: SUPPORTED mean=0.9333333333333333 std=0.24944382578492943 support_fraction=1.0
```

Judge reasoning

All trials used n=1, making the Fourier coefficient sum trivially 1. This violates scaling requirements for both O(log n) and Ω(n) claims. | next: Test with n ≥ 1000 and varying k to observe scaling behavior of Fourier coefficient sums

---

Real Radical Dimension Bounds AC^0 Circuit Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Size for PARITY
- **Recorded**: 2026-05-21 06:31 UTC
- **Entry ID**: `2c022a5d9d33`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the dimension of the real radical of the ideal generated by C's truth-table polynomial equations satisfies $\dim(\mathrm{rad}(I\_C)) \geq \log_2(\mathrm{size}(C)) - O(\sqrt{n})$.

Rationale

Real radicals capture the algebraic structure of boolean functions; their dimension may encode circuit complexity via elimination theory. PARITY's inherent symmetry could create nontrivial constraints on radical varieties, linking algebraic geometry to circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `241.79s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; support fraction cannot be evaluated | next: Run test with extended timeout and verify implementation correctness

---

Eigenvalue Count and SOS Degree for Max-CUT Approximation

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 06:56 UTC
- **Entry ID**: `c0336452dd8d`

Statement

For a random graph G(n, 1/2), the minimal SOS degree required to approximate max-CUT within a factor of 0.878 + ε is Ω(√n), where the constant depends on the number of eigenvalues of G's adjacency matrix in the interval [−√(n log n), √(n log n)].

Rationale

The eigenvalue distribution of random graphs influences the structure of the feasible region for SOS relaxations. Concentration of measure in random matrices suggests that eigenvalues cluster around certain values, which could necessitate higher SOS degrees to capture the necessary precision for approximation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds': Tr

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 71, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 43, in run_trial
    eigenvalues = gaussian_elimination(adjacency_matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_71ec784e.py", line 26, in gaussian_elim
    factor_i = 1 / augmented_matrix[i][i]
               ~~^~~~~~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable evaluation of the conjecture's validity across sufficient instances. | next: Fix the gaussian_elimination function to handle singular matrices, then re-run tests with more seeds to validate the conjecture

---

Prime Density in Arithmetic Progressions and Seed Length of Nisan-Wigderson PRGs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Number Theory × Seed Length of Nisan-Wigderson PRGs
- **Recorded**: 2026-05-21 07:37 UTC
- **Entry ID**: `ffe6d4da52cf`

Statement

For a 3-SAT instance with n variables, the seed length of a Nisan-Wigderson PRG that fools AC^0 circuits is $\Theta(\pi(n))$, where $\pi(n)$ is the number of primes $\leq n$. This holds for all instances where the clause graph is bipartite and the variable assignments induce a Dirichlet progression with modulus $\leq n^2$.

Rationale

The distribution of primes in arithmetic progressions relates to the structure of number fields, which could influence the pseudorandomness properties needed for PRGs. The density of primes affects the entropy required to generate pseudorandom bits, thus influencing seed length.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.06s`


```
': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds': False, 'counterexa
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
TRIAL: {'metric_name': 'Seed Length', 'metric_value': None, 'instances_tested': 0, 'conjecture_holds
RESULT: FALSIFIED counterexample="Modulus must be a prime number" first_failing_seed=11
```


Judge reasoning

No instances were tested, violating the 'n too small' failure mode. The counterexample references modulus requirements but lacks empirical validation. | next: Run tests with bipartite clause graphs and prime moduli to collect empirical data on seed length behavior

---

Free Entropy Gap in Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Communication Complexity of Disjointness
- **Recorded**: 2026-05-21 07:51 UTC
- **Entry ID**: `ba26a4ca7784`

Statement

For a random Disjointness matrix M ∈ {0,1}^{n×n}, let $\chi(M)$ denote the free entropy of its entries treated as non-commutative random variables. Then $\chi(M) \geq \Omega(\sqrt{n})$ holds with probability ≥ 0.95 over random seeds, and $\chi(M) = \Theta(\sqrt{n})$ implies the randomized communication complexity of M is $\Omega(n)$.

Rationale

Free entropy quantifies the 'non-commutative complexity' of matrix entries, capturing dependencies that correlate with communication bottlenecks. By framing Disjointness matrices as non-commutative distributions, we expose structural constraints that may lower-bound communication requirements via free probability's asymptotic freeness properties.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
lue': -0.5625, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'chi(M) < 4√n'}
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.625, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.4375, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5625, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.375, 'instances_tested': 1, 'conjecture_ho
TRIAL: {'metric_name': 'free_entropy', 'metric_value': -0.5, 'instances_tested': 1, 'conjecture_hold

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <module>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a045ef7b.py", line 60, in <genexpr>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                                             ~~~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing validation; counterexamples show failure but cannot trust results due to crash | next: Fix seed handling in test code and re-run with sufficient sample size

---

SOS Moment Matrix Spectral Gap and Max-CUT Approximation Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 08:09 UTC
- **Entry ID**: `49361c02c035`

Statement

For every Max-CUT instance with n variables, if the degree-d SOS moment matrix M_d has minimal eigenvalue λ_min(M_d) ≥ ε, then the integrality gap of the d-round SOS relaxation is ≤ 1 - ε. Conversely, if the integrality gap exceeds 0.878, then λ_min(M_d) < 1 - 0.878 for all d ≤ log n.

Rationale

The spectral gap of the moment matrix encodes the SOS relaxation's ability to capture low-degree certificates. By linking eigenvalue thresholds to approximation ratios, we expose structural constraints between algebraic geometry and optimization hardness, avoiding algebrization barriers through explicit spectral analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.87s`

```
value': 1.0142857142857142, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0714285714285714, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0059751773049646, 'instances_tested': 3,
epsilon=525/188'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0130870210295446, 'instances_tested': 3,
epsilon=23245/5392'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0416666666666667, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.004379479878373, 'instances_tested': 3,
epsilon=131545706/32251283'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.01875, 'instances_tested': 3, 'conjecture
epsilon=13/4'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0042481778622288, 'instances_tested': 3,
epsilon=16057508/4367783'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0113157894736842, 'instances_tested': 3,
epsilon=587/200'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0110294117647058, 'instances_tested': 3,
epsilon=5/2'}
RESULT: FALSIFIED counterexample="Gap exceeds 1 - epsilon" first_faili
```

Judge reasoning

parse_fail | next: NONE

Additive Energy of Fourier Coefficients and ACC⁰ Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC⁰ Circuit Size
- **Recorded**: 2026-05-21 08:19 UTC
- **Entry ID**: `2088c7f83a48`

Statement

For every Boolean function f: {0,1}^n → {0,1} in P, the additive energy E(f) of its Fourier coefficients satisfies E(f) ≥ 2^{Ω(n)}, whereas for ACC⁰ circuits computing f, E(f) ≤ $2^{\{O(\log}$2 n)}.

Rationale

Additive energy captures the structure of Fourier coefficients, which reflect the correlation of f with low-degree polynomials. ACC^0 circuits, constrained by bounded-depth, exhibit limited structure, leading to lower additive energy. This could expose a gap between P and ACC^0 via Fourier-analytic methods.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 47, in run_trial
    transformed_f = fast_walsh_hadamard_transform(f)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1fe85aa.py", line 26, in fast_walsh_ha
    v = f[j + m]
        ~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during execution, preventing evaluation of the conjecture's validity. | next: Debug the fast_walsh_hadamard_transform implementation to fix index out of range error

---

Betti Number Sum Lower Bound for SOS Max-CUT Degree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology × SOS Degree for Max-CUT
- **Recorded**: 2026-05-21 08:33 UTC
- **Entry ID**: `b782e0cb2c69`

Statement

For a graph G on n vertices, the SOS degree required to approximate Max-CUT with 0.878-approximation ratio is at least the sum of the Betti numbers $\beta_0 + \beta_1$ of G's clique complex. This bound is tight for complete graphs.

Rationale

Betti numbers encode topological complexity of the clique complex, which may correlate with the algebraic structure of the Max-CUT instance. The SOS hierarchy's degree reflects the need to resolve higher-dimensional interactions, potentially mirroring the topological complexity captured by Betti numbers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s


```
e': ''}
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'SOS Degree', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': Tr
RESULT: SUPPORTED mean=1.0666666666666667 std=0.24944382578492943 support_fraction=1.0
```


Judge reasoning

All trials used n=1 graphs where Betti numbers are trivial, making the conjecture vacuous. SOS degree is capped at 1, preventing meaningful validation of the lower bound. | next: Test conjecture on graphs with n ≥ 4 to observe non-trivial Betti number interactions

---

Asymptotic Dimension and Tseitin Resolution Length

- **Verdict**: INCONCLUSIVE
- **Bridge**: Geometric Group Theory × Resolution Length of Tseitin Formulas
- **Recorded**: 2026-05-21 08:47 UTC
- **Entry ID**: 2f7cf0867b85

Statement

For a graph G with asymptotic dimension d, the resolution length of the Tseitin formula on G is at least 2^{Ω(d)}.

Rationale

Asymptotic dimension captures the large-scale geometric complexity of the graph, which may directly influence the length of resolution proofs required to refute the Tseitin formula.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 117, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 96, in run_trial
    literals, clauses = tseitin_formula(graph)
                        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_98637aa2.py", line 41, in tseitin_formu
    clauses.append([-literals[2*i], literals[2*j]])
                                    ~~~~~~~~^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing valid results | next: Fix the Tseitin formula implementation to handle graph vertex indexing properly

---

Specht Block Spread of Perm Exceeds Det Under Random Linear Lift

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic combinatorics: Specht-module projections via Murnaghan-Nakayama characters on the regular S_n-representation × GL(n^2)-orbit spread of perm_n vs det_n in matching-coefficient basis (Mulmuley-Sohoni GCT, monotone setting)
- **Recorded**: 2026-05-21 09:38 UTC
- **Entry ID**: `48b7235cd4ff`

Statement

Let X=(x_{ij}) be an n×n matrix of formal variables. For a multilinear degree-n polynomial g in x_{ij}, define the matching coefficient vector c(g)∈Q^{S_n} by c_σ(g)=[∏*i x*{i,σ(i)}] g. For each λ⊢n, let P_λ be the projection onto the λ-isotypic component of Q[S_n] using the central idempotent e_λ=(f^λ/n!)∑_σ χ_λ(σ)σ. Define the Specht block count ρ(g)=#{λ⊢n: ||P_λ(c(g))||²/||c(g)||² > 10^{-9}}. Conjecture: ρ(perm_n)=ρ(det_n)=1 for all n≥2; and for every n∈{4,5,6}, for a uniformly random invertible L∈{-1,0,+1}^{n²×n²}, with probability ≥1-1/n we have ρ(perm_n(L(X))) ≥ ρ(det_n(L(X)))+1.

Rationale

Perm_n's matching vector is the all-ones (max-entropy) direction in Q^{S_n}; det_n's is the sign vector — both extremal but on opposite 1-dim irreps. A random linear lift L (the GCT analog of a LIFTING query-to-communication gadget) breaks the S_n×S_n grading that pins each to a single Specht block. Under random L, the trivial-rep-anchored perm should spread across STRICTLY MORE Specht blocks than the sign-rep-anchored det, because the sign vector's interaction with the regular representation under generic linear lift is more rigid (a Schur-Weyl asymmetry inherited from the Foata bijection inverting maj-inv).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 189, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 126, in run_trial
    rho_perm_n = sum(projection_on_irrep(identity_matrix(n), lambda_partition=[1]*n) > 1e-
9 for lambda_partition in partitions(n)) / len(partitions(n))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 126, in <genexpr>
    rho_perm_n = sum(projection_on_irrep(identity_matrix(n), lambda_partition=[1]*n) > 1e-
9 for lambda_partition in partitions(n)) / len(partitions(n))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 79, in projection_on
    f_lambda = Fraction(factorial(n), factorial(lambda_partition))
                                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4e5c65c.py", line 22, in factorial
    for i in range(2, n + 1):
                      ~~^~~
TypeError: can only concatenate list (not "int") to list
```

Judge reasoning

The test crashed with a TypeError before producing any data (factorial was called on a list instead of an integer), so neither the sanity check nor the main 30-seed trial was evaluated. | next: Fix the bug by computing the hook-length formula for f^λ from the partition (or pass n=sum(λ)) and rerun the sanity check for n∈{4,5} before launching the random-lift trials.

Specht Character Support Bounds Monotone Permanental Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Specht modules and Murnaghan-Nakayama character theory of the symmetric group S_n × Monotone arithmetic formula size for multilinear permanental polynomials (GCT det-vs-perm monotone setting)
- **Recorded**: 2026-05-21 09:45 UTC
- **Entry ID**: `b5cefcac0023`

Statement

For a multilinear permanental polynomial f = Σ_{σ∈S_n} c_f(σ) Π_i x_{i,σ(i)} in n×n matrix variables with c_f ≥ 0, define Specht coefficients α_λ(f) := (1/n!) Σ_σ c_f(σ) χ_λ(σ) for each partition λ ⊢ n, where χ_λ is the irreducible character (computed via Murnaghan-Nakayama), and the effective Specht support |supp_eff(f)| := (Σ_λ (dim V_λ · α_λ)$^{2)}$2 / Σ_λ (dim V_λ · α_λ)^4 (inverse participation ratio). Conjecture: every monotone permanental arithmetic formula F of size s (binary tree of ⊕ gates and ⊗ gates that multiply polynomials in disjoint row- AND column-subsets, with nonnegative leaf weights) computing f satisfies |supp_eff(f)| ≤ s; consequently any ‘pure’ polynomial like perm_n (|supp_eff|=1, character (n)) or its sign-twin det_n (character (1^n)) requires monotone permanental size ≥ 1 trivially, but every monotone formula F that produces a polynomial f with |supp_eff(f)|=k must have size ≥ k. Falsifier: a single monotone permanental formula F with size s < |supp_eff(f_F)|.

Rationale

Mulmuley-Sohoni view perm vs det through representation theory of GL-orbit closures; we drop to the more elementary diagonal S_n action where Murnaghan-Nakayama gives O(n! · p(n))-time exact computation of α_λ, bypassing both the natural-proofs barrier (the invariant is relational/categorical — a participation ratio across irreducible isotypics, not a single truth-table number) and algebrization (the construction is not a polynomial-ring identity in the boolean variables but a representation-theoretic projection on S_n cosets). The bound |supp_eff(f)| ≤ s reflects the intuition that each monotone gate can only construct one new ‘pure character direction’, so a formula must have at least as many gates as the character spread of its output — a quantitative refinement of the Jerrum-Snir style monotone obstruction.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:3232ed274e19727bb02719b0038eeab0dad19] Stability in Topology, Arithmetic, and Representation Theory 2022 - [s2:a386ca523f19c66f4e550fdeff35ca56408af] STACS 2006, 23rd Annual Symposium on Theoretical Aspects of Computer Science, Marseille, France, February 23-25, 2006, P

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 148, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 119, in run_trial
    f, chi_table = generate_random_formula(n, s)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2ae01e1a.py", line 107, in generate_ran
   chi_table = {lam: irreducible_character(lam, beta) for lam in itertools.combinations(range(n), n /
                                                ^^^^
NameError: name 'beta' is not defined
```

Judge reasoning

The test crashed with a NameError (‘beta’ is not defined) before producing any data, so the pre-registered support condition cannot be evaluated. | next: Fix the bug in generate_random_formula (define/pass ‘beta’ to irreducible_character) and rerun the full 720-trial sweep.

Möbius Mass of Rectangle Poset Lower-Bounds D^cc

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Möbius-function theory / Rota’s incidence-algebra of finite posets × Deterministic two-party communication complexity D^cc of explicit Boolean functions
- **Recorded**: 2026-05-21 09:52 UTC
- **Entry ID**: `55903661e64a`

Statement

Let $f:\{0,1\}^{n\times\{0,1\}}n\to\{0,1\}$ and let M_f be its $2^{n\times 2}n$ communication matrix. Define P_f as the finite poset whose elements are the maximal 1-monochromatic combinatorial rectangles R=A×B of M_f, ordered by componentwise inclusion (A×B ≤ A’×B’ iff A⊆A’ and B⊆B’), and let μ_f be the Möbius function on its incidence algebra. Conjecture: for every f with D^cc(f) ≥ 4, the total Möbius mass MM(f) := Σ_{R≤R’ in P_f} |μ_f(R,R’)| satisfies log_2 MM(f) ≥ D^cc(f)/4; a single f with D^cc(f) ≥ 4 and log_2 MM(f) < D^cc(f)/4 falsifies it.

Rationale

D^cc(f) is governed entirely by the lattice of monochromatic rectangles, but standard lower bounds use only coarse cardinality features (cover/partition/fooling). The Möbius function detects how far this lattice is from a Boolean algebra: hard functions should generate many genuinely non-distributive intervals among maximal rectangles, producing exponential signed cancellation that mirrors the protocol-tree branching needed to separate them. Because MM is a relational invariant of the rectangle poset (not a single truth-table number) and is not preserved by ring/polynomial extension of the matrix entries, it avoids natural-proofs single-number traps and the algebrization route.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c3ade04.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c3ade04.py", line 24, in run_trial
    M = communication_matrix(f)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1c3ade04.py", line 40, in communication
    n = len(next(iter(func.values())))
                      ^^^^^^^^^^^
AttributeError: 'function' object has no attribute 'values'
```

Judge reasoning

The test crashed with an AttributeError (the code called .values() on a function object) before any trials produced data, so neither the support nor falsification condition can be evaluated. | next: Fix communication_matrix to accept a callable f:$\{0,1\}^{n\times\{0,1\}}$n→{0,1} (infer n from the signature or pass n explicitly) and rerun the pre-registered sweep.

Discrete Morse Number of Rejection Complex Bounds ACC^0[m] Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Morse theory (Forman 1998; Benedetti–Lutz randomized greedy discrete Morse algorithm) × Depth-d ACC^0[m] circuit size for monotone Boolean functions
- **Recorded**: 2026-05-21 09:59 UTC
- **Entry ID**: `911c1d7528b8`

Statement

For a monotone Boolean function f: {0,1}^n → {0,1}, define the rejection complex K(f) := {S ⊆ [n] : f(1_S) = 0}, which is a downward-closed abstract simplicial complex on [n] by monotonicity. Let μ(f) be the number of critical cells output by Benedetti–Lutz's randomized greedy discrete Morse matching on the Hasse diagram of K(f), reported as the median over 30 random seeds. Conjecture: for every fixed constant depth d and prime power m≥2, every depth-d ACC^0[m] circuit (not required to be monotone) computing f has size at least 2^{Ω((log μ(f))^{1/d})}; equivalently, if such a circuit has size s, then μ(f) ≤ 2^{O((log s)^d)}.

Rationale

Discrete Morse theory supplies a relational, Hasse-diagram-level topological invariant — pairings of faces with cofaces — that to our knowledge has zero prior application to ACC^0 lower bounds, and is structurally different from the truth-table single-number invariants (Mahler measure, magnitude, doubling) that have repeatedly triggered NATURAL_PROOFS. Critical cells encode unavoidable homological obstructions to collapse, which the Razborov–Smolensky polynomial-approximation core of Williams's NEXP⊄ACC^0 program suggests must be absent for functions admitting low-degree F_m approximators. The mapping uses only face incidences in 2[1], not arithmetic in a polynomial ring extension of {0,1}, so the construction does not algebrize in the Aaronson–Wigderson sense.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 139, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 119, in run_trial
    mu = run_benedetti_lutz(f_instance, n)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 110, in run_benedett
    if f(S) == 0:
       ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 95, in <lambda>
   return lambda x: any(all(x[i] == c for i, c in enumerate(clause)) for clause in clauses)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 95, in <genexpr>
   return lambda x: any(all(x[i] == c for i, c in enumerate(clause)) for clause in clauses)
                            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba9404da.py", line 95, in <genexpr>
   return lambda x: any(all(x[i] == c for i, c in enumerate(clause)) for clause in clauses)
                                ~^^^
TypeError: 'int' object is not subscriptable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not subscriptable) inside the function evaluator before any μ data was collected, so no support condition could be evaluated. | next: Fix the f_instance construction so that inputs S are converted to a proper indicator vector x (e.g., x = [1 if i in S else 0 for i in range(n)]) before evaluating clauses, then rerun across n∈{8,10,12,14} with 30 seeds.

Trace-Norm Capacity Conjecture for Read-Twice BP Communication Matrices

- **Verdict**: `INCONCLUSIVE`

[1] n

- **Bridge**: Schatten-1 (nuclear/trace) norm geometry of sign matrices in the Linial–Shraibman γ_2/discrepancy-lifting tradition × Read-twice oblivious branching programs (Borodin–Razborov–Smolensky model), measured via their balanced communication matrix
- **Recorded**: 2026-05-21 10:06 UTC
- **Entry ID**: `0bd071f00973`

Statement

Let P be a read-twice oblivious branching program of size s on n input bits (n even) computing f_P:{0,1}^n→{±1}. For the canonical balanced bipartition into the first n/2 and last n/2 input variables, form the $2^{\{n/2\}\times 2}${n/2} sign-matrix M_P[x,y]=f_P(x,y) and let ‖M_P‖* *denote its nuclear (sum of singular values) norm. Define ρ(P):=log_2(‖M_P‖*/2^{n/2}); we conjecture there is an absolute constant C≤4 such that ρ(P)≤C·log_2(s+1) for every read-twice oblivious BP P, while the trivial truth-table BP for IP_2 (which has size 2^n) attains ρ=n/4=Ω(n), so any read-twice BP computing IP_2 must have size ≥2^{n/(4C)}.

Rationale

For read-once oblivious BPs the communication matrix has real rank ≤width(P), instantly giving ρ≤log width, but this rank argument breaks at the second read — exactly the obstruction the Borodin–Razborov–Smolensky 1993 program ran into. Lifting BP size to a Schatten-1 norm (rather than rank) is a natural relaxation that survives shared layer reuse, and a polynomial trace-norm bound would be a clean operator-theoretic lift of BP size, automatically converting the Lindsey-lemma discrepancy 2^{-n/2} of IP_2 into an exponential read-twice lower bound. Because nuclear norm is convex and submultiplicative under layer products, it is a plausible candidate where free-probability / spectral attempts on this problem repeatedly stalled.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [0809.3137v1] The Frontiers of Nuclear Science, A Long Range Plan - [0809.2093v2] An approximation algorithm for approximation rank - [1901.06841v1] Consistency Examinations of Calculations of Nuclear Matrix Elements of Double-β Decay by QRPA

Empirical Test

- exit code: `1`, elapsed: `0.20s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47d72295.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47d72295.py", line 62, in run_trial
    rho_P = math.log2(nuclear_norm(M) / 2**(n//2))
                      ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47d72295.py", line 49, in nuclear_norm
    U, _, Vt = gaussian_elimination(matrix_multiply(M, M))
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_47d72295.py", line 31, in gaussian_elim
    A[i][j] /= pivot
ZeroDivisionError: float division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in a hand-rolled gaussian_elimination routine used to compute the nuclear norm, so no data was produced for either the random BP instances or the IP_2 baseline. Neither side of the pre-registered support condition could be evaluated. | next: Replace the buggy custom SVD with numpy.linalg.svd (nuclear norm = sum of singular values) and re-run the 450 random instances plus the IP_2 truth-table baseline across n∈{8,10,12,14,16}.

Slice-Fourier Variance Asymmetry of Perm vs Monotone Det Substitutions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis on Boolean-cube slices / Johnson-scheme harmonic decomposition (Filmus–O'Donnell slice-Fourier theory) × GCT-monotone permanent-vs-determinant: monotone padded det_m linear substitutions targeting perm_n
- **Recorded**: 2026-05-21 10:14 UTC
- **Entry ID**: `bbd70a9083de`

Statement

For X uniform on the slice S_n ⊂ {-1,+1}^{n²} of Hamming weight n (exactly n entries equal −1), define μ(f) := E_{X~S_n}[f(X)²] and the slice-asymmetry ξ(f) := (μ(perm_n) − μ(f)) / (μ(perm_n) + μ(f)). Conjecture: (i) ξ(det_n) ≥ 1 − 4^{−n} for all n ≥ 2; and (ii) for every nonnegative linear substitution g(X) := det_m(BX) with B ∈ R_{≥0}^{m² × n²} and m ≤ ⌊n^{3/2}⌋, the asymmetry satisfies ξ(g) ≥ 1 − n·4^{−n}. A single (n, m, B) with n ≤ 6 violating either bound refutes the conjecture.

Rationale

Perm_n and det_n share monomial support and identical L²-mass on the full ±1 cube, so global Fourier-spectral invariants cannot separate them, but on the low-weight slice S_n the harmonic projection onto Schur components forces det's signed expansion to cancel destructively against the slice geometry while perm's all-positive expansion adds constructively — giving an exponentially large ξ. Restricting the substitution matrix B to be entrywise nonnegative preserves perm's coefficient-positivity advantage even after linear reparameterization, turning the slice asymmetry into a constructive Fourier-analytic monotone invariant for the GCT det-vs-perm program. The invariant is computed via explicit slice-harmonic projections (signed sums of evaluations), not via a poly-time predicate on a truth table, so it evades the Razborov–Rudich natural-proofs trap targeted by hypercontractive influence pioneers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f3aee405.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f3aee405.py", line 36, in run_trial
    if xi(det_n, 2) < 1 - 4**-2:
       ^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f3aee405.py", line 32, in xi
    return (mu(perm_n, n) - mu(f, n)) / (mu(perm_n, n) + mu(f, n))
            ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f3aee405.py", line 27, in mu
```

```
    total += f(X)**2
             ^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f3aee405.py", line 31, in <lambda>
    perm_n = lambda X: sum(X[i*n:(i+1)*n] for i in range(n))
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: unsupported operand type(s) for +: 'int' and 'list'
```

Judge reasoning

The test crashed with a TypeError in the perm_n lambda (summing integers with list slices) before any data was produced, so neither the support nor falsification condition was evaluated. | next: Fix the perm_n implementation to compute the matrix permanent correctly (e.g., via Ryser's formula on the reshaped n×n matrix) and rerun the pre-registered protocol for n∈{2,3,4,5}.

VC dimension of gadget row family tightly bounds lifted D^cc

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Vapnik-Chervonenkis combinatorial dimension theory × Query-to-communication lifting for deterministic two-party protocols (Raz-McKenzie gadget composition)
- **Recorded**: 2026-05-21 10:22 UTC
- **Entry ID**: `02a2f4e161dc`

Statement

Let g: A × B → {0,1} be a gadget and let R_g = {g(a,·) : a ∈ A} ⊆ 2^B be its row family, with VC dimension d = VC(R_g). For every Boolean function f : {0,1}^n → {0,1} with deterministic decision-tree depth D(f) ≥ 2, the deterministic communication complexity of the Raz-McKenzie lift satisfies D^cc(f ∘ g^n) ≥ (1/2)·D(f)·d − n. Equivalently, log_2 rank_{GF(2)}(M_{f∘g^n}) ≥ (1/2)·D(f)·VC(R_g) − n for every (f,g).

Rationale

Standard lifting theorems give the multiplier log b for index_b, where b is also exactly VC(R_{index_b}); for inner-product gadgets the same equality holds. We conjecture that across ALL gadgets the per-coordinate information cost of distinguishing rows in any protocol is governed not by |A| or by rank but by the shatter dimension of R_g, because every protocol leaf must combinatorially separate a shattered family. This bridges PAC-style combinatorial dimension with the structural side of lifting and yields a falsifier if any low-VC gadget achieves anomalously high D^cc.

Novelty

- Judge: `NOVEL` over 7 arXiv hits

Top hits consulted: - [1503.07648v2] Sign rank versus VC dimension - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2001.02144v1] Lifting with Simple Gadgets and Applications to Circuit and Proof Complexity - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3e622c.py", line 93, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ad3e622c.py", line 66, in run_trial
    M = [[f[int(''.join(map(str, g(a, b))), 2)] for b in range(B_size)] for _ in range(A_size**n)]
                                      ^
UnboundLocalError: cannot access local variable 'a' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so no support fraction or slack statistics were computed. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the bug in the matrix construction (the inner comprehension iterates over '_' instead of binding 'a', so 'a' is unbound); enumerate $a \in A^n$ properly via itertools.product, then rerun the 30-seed protocol.

Gromov 4-Point Hyperbolicity Gap Separates k-CLIQUE Monotone DNFs from Random

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse geometry — Gromov δ-hyperbolicity of finite metric spaces (rarely applied to monotone complexity; appears mostly in geometric group theory and network science) × Monotone DNF representations of the k-CLIQUE indicator (Razborov monotone lower-bound setting, viewed as a query-to-DNF lifting target)
- **Recorded**: 2026-05-21 10:32 UTC
- **Entry ID**: `0420f93e5160`

Statement

For a monotone k-uniform DNF $F = T_1 \vee \dots \vee T_m$ on n Boolean variables (variables = edges of K_v where $n = C(v,2)$) with supports $S_i \subseteq [n]$ of size k, define the symmetric-difference metric $d_F(i,j) = |S_i \triangle S_j|$ and the normalized 4-point Gromov hyperbolicity $\delta(F) = \max_{\{i,j,k,l\}} ((M_1 - M_2)/(2 \cdot \mathrm{diam}(F)))$ *where* $M_1 \geq M_2 \geq M_3$ *are the sorted values of* $\{d(i,j)+d(k,l), d(i,k)+d(j,l), d(i,l)+d(j,k)\}$*. CONJECTURE: (i) For the CANONICAL k-CLIQUE DNF* F_{clique} *whose clauses enumerate every k-clique of* K_v ($k = \lfloor\sqrt{v}\rfloor$), $\delta(F_{clique}) \geq 1/4 - o(1)$ as $v \to \infty$; (ii) For a uniformly random k-uniform monotone DNF F_{rand} with the same n, k, and number of clauses, $E[\delta^*(F_{rand})] \leq C \cdot v^{-1/2} \cdot \log v$ for an absolute constant C. Hence the ratio $\delta^*(F_{clique})/E[\delta^*(F_{rand})]$ grows as $\Omega(\sqrt{v} / \log v)$, giving a structural separator that submodularly lower-bounds any monotone DNF refinement of k-CLIQUE.

Rationale

Gromov hyperbolicity quantifies how 'tree-like' a metric is on four-point configurations; the k-clique family in K_v has a rigid Johnson-scheme metric where almost every 4-tuple has two near-equal large pairwise sums (witnessing high δ*), while uniformly drawn k-subsets are with high probability metric-quasi-orthogonal (δ* concentrates near 0 by classical Hamming-cube concentration). If the gap persists, it identifies a coarse-geometric obstacle that any small monotone DNF cover of CLIQUE must encode, providing a query-like invariant ripe for Raz–McKenzie-style lifting into monotone circuit lower bounds without invoking sum-check or polynomial extension (so algebrization does not bite).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2311.04204v3] Sharp Thresholds Imply Circuit Lower Bounds: from random 2-SAT to Planted Clique - [1307.4308v2] An Alternative Proof of the Exponential Monotone Complexity of the Clique Function - [2401.17801v2] Weighted-Hamming Metric for Parallel Channels - [0603539v2] Characterizations of metric trees and Gromov hyperbolic spaces

Empirical Test

- exit code: `1`, elapsed: `1.56s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe3adfed.py", line 128, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe3adfed.py", line 75, in run_trial
    delta_F_clique = gromov_hyperbolicity(F_clique, v)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe3adfed.py", line 49, in gromov_hyperb
   d_F[i][j] = abs(sum(1 for k in F if k & (1 << i)) - sum(1 for k in F if k & (1 << j)))
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fe3adfed.py", line 49, in <genexpr>
   d_F[i][j] = abs(sum(1 for k in F if k & (1 << i)) - sum(1 for k in F if k & (1 << j)))
                                        ~~^~~~~~~~~~
TypeError: unsupported operand type(s) for &: 'set' and 'int'
```

Judge reasoning

Test crashed with a TypeError (bitwise & applied to a set instead of an int) before any data was produced, so neither the support nor falsification condition can be evaluated. | next: Fix the clause representation in gromov_hyperbolicity (encode each clause as a bitmask integer or compute d_F via |S_i △ S_j| on the actual support sets) and rerun the pre-registered sweep.

Lorentzian Hessian Gap of Spanning-Tree Polynomial Bounds Max-CUT SOS Slack

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Lorentzian polynomials and combinatorial Hodge theory of matroids (Brändén–Huh 2020; Adiprasito–Huh–Katz), interpreted via Burton–Pemantle transfer currents on uniform random spanning trees × Sum-of-squares degree-2 (Goemans–Williamson / Delorme–Poljak) integrality slack for Max-CUT
- **Recorded**: 2026-05-21 10:48 UTC
- **Entry ID**: `45433c968a7a`

Statement

For a connected graph G=(V,E) with |V|=n, |E|=m, let f_G(x_e)=Σ_{spanning tree T}∏*{e∈T}x_e be its multilinear spanning-tree polynomial; f_G is Lorentzian, so its Hessian H_G∈R^{m×m} at x=1, whose entries are H_G[e,e']=τ*{e,e'} for e≠e' and 0 on the diagonal, has signature (+,−,−,…,−) with a unique positive eigenvalue λ_1(H_G)>0 and remaining eigenvalues λ_2≥…≥λ_m all ≤0. Define the Lorentzian Hessian gap g(G):=λ_1(H_G)−λ_2(H_G)≥λ_1(H_G)>0 (normalized by τ(G) so that off-diagonal entries equal P(e,e'∈T)). CONJECTURE: there exists an absolute constant c>0 such that for every connected G with n≥6 and m≥n, the Delorme–Poljak SOS-2 upper bound DP(G):=m/2+(n/4)|λ_min(A_G)| exceeds the true MaxCut by at least DP(G)−MaxCut(G) ≥ c·g(G)·n; a single graph G violating this for the empirically calibrated c falsifies the conjecture.

Rationale

The Lorentzian Hessian encodes the global negative-correlation structure of edges in a uniform random spanning tree (Burton–Pemantle), an instance-level matroid-Hodge invariant orthogonal to the cut-spectral structure that SOS-2 exploits; a large gap g(G) signals matroid-balanced ‘spread’ that we predict the eigenvalue/SDP relaxation cannot reproduce, forcing slack against true MaxCut. The invariant is real-geometric (Hessian signatures over R), not ring-extendable over a polynomial extension of F_2, so the construction does not algebrize; and the property is read off the graph G as an instance, not from a Boolean truth table, so the natural-proofs largeness/usefulness combination does not apply. Combinatorial Hodge × SOS Max-CUT is essentially untouched (<5 papers).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2206.02759v4] Polynomials with Lorentzian Signature, and Computing Permanents via Hyperbolic Programming - [2304.08399v3] Dually Lorentzian Polynomials - [2204.02097v2] Simulated Annealing is a Polynomial-Time Approximation Scheme for the Minimum Spanning Tree Problem - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut: λ-Extendible Properties Parameterized Above the Poljak-Turzík Bound

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66e7e22f.py", line 126, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66e7e22f.py", line 81, in run_trial
    L_inv = pinv(L)
            ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_66e7e22f.py", line 44, in pinv
    inverse = [[col[i] for col in reduced_matrix] for i in range(n, 2*n)]
                ~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the custom pinv() routine before any trials completed, so no ratios were computed and the pre-registered support criterion cannot be evaluated. | next: Replace the hand-rolled pinv with numpy.linalg.pinv (or fix the row-reduction bounds) and rerun the full 450-instance sweep.

NW-Design Slice Spectral Discrepancy Bound for DISJ

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Nisan-Wigderson combinatorial designs (set families with bounded pairwise intersection, Nisan–Wigderson 1994; Impagliazzo–Wigderson 1997) × Spectral / discrepancy lower bound on the bounded-error randomized communication complexity of DISJOINTNESS (Razborov 1992 setting)
- **Recorded**: 2026-05-21 10:54 UTC
- **Entry ID**: `25b153950ac9`

Statement

For each n, let $D_n = \{S_1,\dots,S_m\}$ be an (n, ℓ, k)-Nisan–Wigderson design with $ℓ = \lceil \log_2 n \rceil$, $k = \lceil \log_2 \log_2 n \rceil$, and $m = m(n) \geq \lfloor n/(2k+1) \rfloor$, constructed by greedy random sampling of ℓ-subsets of [n] subject to $|S_i \cap S_j| \leq k$. Define the design-slice matrix $\Sigma_n(f) \in \{\pm 1\}^{m \times m}$ by $\Sigma_n(f)[i,j] = 1 - 2 \cdot f(\chi(S_i), \chi(S_j))$, and the normalized spectral discrepancy $\rho(f, D_n) := \sigma_{max}(\Sigma_n(f))/m$. Conjecture: $\rho(DISJ_n, D_n) \leq C \cdot m^{-1/2}$ for an absolute constant $C \leq 4$ and every valid NW design D_n with $n \geq 12$, whereas $\rho(EQ_n, D_n) \geq 1 - 6/m$; equivalently, every NW-design slice of DISJ is spectrally as flat as a random sign matrix, certifying $disc(\Sigma_n(DISJ)) \leq C \cdot m^{-1/2}$ and hence $R^{pub}_{1/3}(DISJ_n) \geq \Omega(\log m)$ per design family (constants tracked across families yield $\Omega(n)$ by varying ℓ,k).

Rationale

NW designs are precisely the explicit pseudorandom set families Razborov–Rudich identify as the canonical way to evade the Natural Proofs barrier, so an invariant defined through a design slice is structural rather than a generic truth-table statistic. The slice picks out exactly those input pairs whose AND-support sizes are tightly controlled by $|S_i \cap S_j| \leq k$, so DISJ's entries decorrelate predictably while EQUALITY collapses to a near-identity, mirroring the design's pseudo-independence guarantee. If the bound holds, plugging the discrepancy method gives a constructive, non-natural witness route to the $\Omega(n)$ randomized lower bound for DISJ.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2401.14623v1] Structure in Communication Complexity and Constant-Cost Complexity Classes - [1505.03110v1] Near-optimal bounds on bounded-round quantum communication complexity of disjointness - [0801.3624v3] Multiparty Communication Complexity of Disjointness - [1009.0997v3] Spectral asymptotics for Robin problems with a discontinuous coefficient - [1208.6015v1] The spectral function of a first order elliptic system

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4f067d57.py", line 141, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4f067d57.py", line 116, in run_trial
    rho = spectral_discrepancy(sigma_matrix)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4f067d57.py", line 20, in spectral_disc
    u, _, _ = svd(sigma_matrix)
              ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4f067d57.py", line 34, in svd
   AAT = [[sum(a[i] * b[j] for a, b in zip(row_a, row_b)) for j in range(n)] for row_a in matrix]
                                                  ^^^^^
NameError: name 'row_b' is not defined. Did you mean: 'row_a'?
```

Judge reasoning

Test crashed with a NameError in the custom svd implementation before any trials produced data, so neither support nor falsification can be assessed. | next: Fix the SVD helper (the inner comprehension is missing its `for row_b in matrix` clause) or replace it with numpy.linalg.svd, then rerun the 240-trial sweep.

MST Persistence Entropy Lower-Bounds Log-Rank of Sign Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Topological data analysis — persistent H_0 entropy of single-linkage hierarchical clustering on a finite metric space (Carlsson barcode theory; Edelsbrunner–Harer; Atienza–Gonzalez-Diaz–Soriano-Trigueros persistence-entropy) × Rank / log-rank lower bounds for ±1 sign matrices M_f arising as communication matrices in deterministic and randomized two-party communication complexity (Razborov, Sherstov, Klartag rigidity tradition)
- **Recorded**: 2026-05-21 11:03 UTC
- **Entry ID**: `007b80963b09`

Statement

For every ±1 matrix M of size n×n with 4 ≤ n ≤ 40, let T be the minimum spanning tree of the rows of M under un-normalized Hamming distance, let □_1,...,□_{n−1} be its edge weights, restrict to the multiset P = {□_i : □_i > 0}, set L = ∑*{□∈P} □, and define the persistence entropy PE(M) =* −∑{□∈P} (□/L) · log_2(□/L) (with PE(M) = 0 if P is empty). Then PE(M) ≤ log_2(rank_ℝ(M) + 1); equivalently 2^{PE(M)} ≤ rank_ℝ(M) + 1. A single ±1 matrix witnessing 2^{PE(M)} > rank_ℝ(M) + 1 refutes the conjecture.

Rationale

Persistent H_0 barcodes / MST edge-length spectra are a multiscale fingerprint of how rows of M cluster in Hamming space; n vectors that span an r-dimensional ℝ-subspace can only realise an MST whose edge lengths concentrate on O(r) effective scales, so their Shannon entropy ought to be log-bounded by the rank. Because PE is a metric/topological functional of the row cloud and not a polynomial in the entries, the bridge sidesteps algebrization, and because we lower-bound matrix rank (a communication-specific quantity, not poly-size circuits) it sidesteps Razborov–Rudich naturalness while still feeding directly into log-rank / discrepancy lower bounds. To our knowledge persistent-homology entropy has not been previously deployed against communication-matrix rank, putting the field_A × field_B pair below the <5-paper threshold.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1955174d.py", line 98, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1955174d.py", line 87, in run_trial
    PE = persistence_entropy(M)
         ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1955174d.py", line 54, in persistence_e
    PE = -sum(p * math.log2(p) for p in P)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1955174d.py", line 54, in <genexpr>
    PE = -sum(p * math.log2(p) for p in P)
                  ^^^^^^^^^^^^
ValueError: math domain error
```

Judge reasoning

The test crashed with a ValueError (math domain error) in persistence_entropy due to a log2(p) call with p=0, before producing any RESULT line or trial data. No violation_fraction or worst_slack was computed, so the pre-registered support condition cannot be evaluated. | next: Fix persistence_entropy to filter □>0 strictly (or skip zero-probability terms) and re-run the full 30 seeds × 4 sizes × 5 families grid to obtain violation_fraction and worst_slack.

Instance-Complexity Certification Depth Bounds Monotone DNF Size for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algorithmic instance complexity (Orponen–Ko–Schöning–Watanabe), realized as resource-bounded conditional Kt approximated by greedy DPLL-style minimal-certificate length on a fixed monotone DNF × Monotone DNF formula size for the k-CLIQUE indicator (Razborov approximation-method setting, edges as variables)
- **Recorded**: 2026-05-21 11:12 UTC
- **Entry ID**: `17e6b7435e5a`

Statement

Let F be a monotone DNF over N=C(n,2) edge variables on graphs with n vertices, and let D_k be the uniform distribution on assignments of Hamming weight exactly C(k,2). For x in {0,1}^N define cert(F,x) := the number of variables revealed by a fixed deterministic greedy certifier (which iteratively reveals the variable maximizing the number of live monotone terms it touches, terminating once F(x) is forced) before F(x) is determined; set μ(F) := E_{x~D_k}[cert(F,x)]. Conjecture: (i) for every monotone DNF F of size ≤ N^c with c ≤ 3, μ(F) ≤ c·log2(N)+8; and (ii) for every monotone DNF F that computes the k-CLIQUE indicator at k=⌈√n⌉, μ(F) ≥ ⌊n/4⌋. A single (F,k,n) violating either bound on at least one of 30 seeds refutes the conjecture.

Rationale

Instance complexity is a meta-complexity invariant that, unlike single-number truth-table properties, is defined on (formula, input) pairs and inherits non-largeness from K-complexity, so it sidesteps Razborov–Rudich while still being a hard combinatorial witness; the greedy certificate is its concrete computable proxy. Poly-size monotone DNFs admit cheap certificates (one quickly-falsifying 0 or one fully-set term), giving the O(log N) ceiling; whereas at the clique threshold k≈√n, any monotone DNF computing k-CLIQUE must, in expectation, expose Ω(n) edges before disambiguating clique from non-clique because both planted-clique and locally-cliquey near-misses survive every short partial assignment. Bridging instance complexity to Razborov's monotone setting is rarely attempted (<5 papers) and would furnish a submodular DNF measure directly meeting the focus's (i,ii,iii).

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [1102.2932v2] Applications of Monotone Rank to Complexity Theory - [2211.03396v3] Certificate Games and Consequences for the Classical Adversary Bound - [0209032v3] Complexity Results on DPLL and Resolution

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5c70bf99.py", line 136, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5c70bf99.py", line 118, in run_trial
    F_clique = generate_k_clique_dnf(n, k)
                                        ^
NameError: name 'k' is not defined
```

Judge reasoning

The test crashed with a NameError ('k' is not defined') before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the scoping bug in run_trial so k is passed into generate_k_clique_dnf, then rerun the full 30-seed sweep across n∈{6..10}, k∈{3,4}.

Cyclic Fourier Spread of Term-Vertex Incidence in Monotone k-CLIQUE DNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation theory of cyclic subgroups C_n ≤ S_n via character-Fourier decomposition of term-vertex incidence (monotone-GCT flavor, Ikenmeyer-style combinatorial reduction of Mulmuley-Sohoni symmetries) × Monotone DNF formula size / structure for the k-CLIQUE indicator (Razborov 1985 monotone setting, edges of K_n as variables)
- **Recorded**: 2026-05-21 11:23 UTC
- **Entry ID**: `f4d1273a6af3`

Statement

For a monotone DNF F = T_1 ∨ … ∨ T_m on the edges of K_n, build M(F) ∈ Z^{m×n} with M(F)[i,v] := |{e ∈ T_i : v ∈ e}|, and let M̂(F)[i,j] := Σ_v M(F)[i,v]·exp(2πi·jv/n) be its column-wise C_n-DFT. Define the cyclic-Fourier spread μ(F) := #{ j ∈ {0,…,n−1} : ∃ i with |M̂(F)[i,j]| > 0 }. Conjecture: for every monotone DNF F that computes the k-CLIQUE indicator on K_n with n ≥ 2k and k ≥ 3, μ(F) ≥ ⌈n/2⌉; equivalently, no monotone DNF computing k-CLIQUE can confine its term-vertex incidence to fewer than half of the cyclic frequencies.

Rationale

In the GCT-monotone program one seeks symmetry-based obstructions: any formula for a highly symmetric function like the k-CLIQUE indicator must, after restricting the S_n action to a tractable cyclic subgroup C_n, exhibit a wide spread across irreducible C_n-characters because the k-subset orbit-structure couples to all nontrivial frequencies. A poly-size monotone DNF that 'collapses' incidence into few C_n-characters would correspond to a forbidden orbit-closure containment for the k-CLIQUE polynomial. Because μ is a complex-character invariant tied to roots of unity (not a polynomial-ring identity), it does not algebrize, and because it depends on the formula F (not the truth table) it dodges natural proofs.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0509044v2] Decomposing symmetric powers of certain modular representations of cyclic groups - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [0902.2349v1]

Defect of characters of the symmetric group - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group

Empirical Test

- exit code: 1, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf4554a.py", line 132, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf4554a.py", line 105, in run_trial
    mu_canonical = cyclic_fourier_spread(canonical_dnf, n)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf4554a.py", line 94, in cyclic_fourie
    M_hat = dft(M, n)
            ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_adf4554a.py", line 87, in dft
    sum_val += M[i][v] * math.cos(2 * math.pi * j * v / n)
               ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the DFT routine before producing any μ(F) measurements, so neither the support nor the falsification criterion could be evaluated. | next: Fix the indexing bug in cyclic_fourier_spread/dft (ensure M is built as m×n with vertices indexed 0..n-1) and rerun the 7×30 trial battery.

Coboundary Spectral Gap Lower-Bounds DPLL Tree Size on Tseitin

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Linial–Meshulam–Wallach high-dimensional simplicial coboundary expansion (spectral gap of the up-Laplacian on 1-cochains) × DPLL backtrack count (tree-like resolution refutation size) for Tseitin formulas on 3-regular graphs
- **Recorded**: 2026-05-21 11:29 UTC
- **Entry ID**: `5ca5616545f4`

Statement

Let G be a connected 3-regular graph on n vertices and c:V(G)→{0,1} have odd Hamming weight, so Ts(G,c) is unsatisfiable. Let X(G) be the 2-complex whose 0-simplices are edges of G, 1-simplices are pairs of G-edges sharing a vertex, and 2-simplices are the (unique) triples of G-edges meeting at each vertex, and let λ↑(G) be the smallest nonzero eigenvalue of the real 1-up-Laplacian δ1* δ1 on C^1(X(G); R). Then there is an absolute constant κ>0 such that for every such (G,c), log2(min DPLL backtracks to refute Ts(G,c)) ≥ κ · λ↑(G) · √n; a single (G,c) violating this bound by ≥20% at n≥18 falsifies the conjecture.

Rationale

Graph-expansion lower bounds on Tseitin (Ben-Sasson–Wigderson) capture only 1-dimensional vertex flow, but DPLL must propagate parity contradictions through local 'around-a-vertex' triples — exactly the 2-simplices of X(G); the Linial–Meshulam up-Laplacian eigenvalue is the canonical spectral surrogate for higher-order Cheeger expansion of that triple-structure. If true, λ↑ provides a polynomial-time computable structural certificate, complementary to vertex expansion, for tree-resolution hardness, and being a property of the formula's incidence geometry (not its truth table) it sidesteps natural proofs, relativization, and algebrization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aee00bc8.py", line 144, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aee00bc8.py", line 111, in run_trial
    lambda_up = compute_lambda_up(X_G)
                ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_aee00bc8.py", line 72, in compute_lambd
   L1 = [[D[2][i] - D[1][j] if (i // 3 == j // 3 and i % 3 != j % 3) else 0 for j in range(m)] for i in rang
           ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in compute_lambda_up before any instances were evaluated, so no data was produced to assess the pre-registered support or falsification conditions. | next: Fix the L1 up-Laplacian construction (correct indexing/boundary maps on the 2-complex X(G)) and re-run the 210-instance protocol.

Heilmann-Lieb Matching Root Gap Bounds Monotone k-CLIQUE DNF Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Statistical mechanics — Heilmann-Lieb monomer-dimer theory: real-rooted matching polynomial of bipartite incidence graphs (Heilmann–Lieb 1972; Godsil identity; Mohar bounds). Rarely applied to monotone circuit/DNF lower bounds (<5 papers). × Monotone DNF formula size for the k-CLIQUE indicator (Razborov 1985 monotone setting; edges of K_n as variables; target separation: super-polynomial monotone DNF lower bound).
- **Recorded**: 2026-05-21 11:40 UTC
- **Entry ID**: `4b37abf22009`

Statement

For a monotone DNF F = T_1 ∨ … ∨ T_m over the edge set E of K_n, let G(F) be the bipartite term–variable incidence graph (one side = the m terms, other side = the |E| edges, an edge of G(F) joining T_i to e iff e ∈ T_i) and let M_F(z) = Σ_k (−1)^k μ_k(G(F)) z^{m+|E|−2k} be its matching polynomial, whose roots are real by Heilmann–Lieb. Define ρ(F) := min{|r| : M_F(r) = 0} and μ(F) := log_2(1 + ρ(F)^2). Conjecture: there exist absolute constants c_1, c_2 > 0 with c_1 < c_2 such that (a) for every monotone DNF F of size m ≤ n^{c_1} computing any monotone Boolean function, μ(F) ≤ c_2 · log_2 n, while (b) for every monotone DNF F whose terms equal exactly the edge-sets of the k-cliques of K_n with k = ⌊log_2 n⌋, μ(F) ≥ √n / c_2 — a single counterexample on either side falsifies the conjecture.

Rationale

The matching polynomial of a bipartite incidence graph encodes how concentrated or spread the term–variable incidence pattern is, and Heilmann–Lieb forces all of its roots onto the real line so the smallest |root| is a clean, robust spectral invariant. A monotone DNF representing k-CLIQUE must reuse each edge across exponentially many heavily overlapping clique-terms, which pushes the bipartite incidence graph toward an essentially regular, high-permanent structure whose matching roots are bounded well away from 0. Small poly-size DNFs over n variables produce sparse incidence graphs whose smallest |root| is forced near 1 by elementary Heilmann–Lieb bounds, so a separation here would yield a structural monotone lower bound technique unconnected to Razborov's approximator method.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5733543c.py", line 141, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5733543c.py", line 87, in run_trial
    random_mu = mu(random_dnf)
                ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5733543c.py", line 76, in mu
    rho = min_root(poly)
          ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5733543c.py", line 61, in min_root
    s = sturm_sequence(poly)
        ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5733543c.py", line 56, in sturm_sequenc
    r = [s[-2][-i-1] - (q[-1][-i-1] * s[-1][-i]) for i in range(len(s[-2])-len(q[-1]))]
                                                                     ~^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the Sturm sequence routine before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the sturm_sequence implementation (correct the index arithmetic in the polynomial remainder step) or replace it with numpy.roots on the matching polynomial coefficients, then re-run.

Fourier L1 Norm Cubed-Polynomial Bound for Read-Twice BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis on the Boolean cube — spectral (Fourier-ℓ¹ / μ-) norm in the Kahn-Kalai-Linial / Bruck-Smolensky tradition × Read-twice oblivious branching programs (BP_READTWICE), Borodin-Razborov-Smolensky 1993 setting; size is total node count
- **Recorded**: 2026-05-21 11:54 UTC
- **Entry ID**: `90dea97bbe27`

Statement

Let P be a read-twice oblivious branching program of size s = size(P) computing f : {0,1}^n → {-1,+1}. Then its Fourier ℓ^1 norm satisfies $\|\hat{f}\|_1$:= Σ_{S⊆[n]} $|\hat{f}(S)|$ ≤ (s+1)^3. As a corollary, since IP_2 on 2n variables has spectrum $\hat{f}(S,T) = (-1)^{\{|S\cap T|\}\cdot 2}${-n} for every (S,T) and hence $\|\hat{IP}_2\|_1$ = 2^n, every read-twice oblivious BP for IP_2 has size at least 2^{n/3} - 1.

Rationale

The Fourier ℓ^1 norm controls approximate junta degree, sign-rank, communication complexity, and PAC-learnability, yet has barely been linked to branching-program size beyond the read-once width-w bound. Crucially the property $\|\hat{f}\|_1$ ≤ poly(s) is a SMALLNESS property — random Boolean functions saturate Cauchy-Schwarz with $\|\hat{f}\|_1$ ≈ 2^{n/2} — so it cleanly evades the Razborov-Rudich largeness condition that killed previous tail-mass attempts. The conjecture asserts a structural Bruck-Smolensky-style spectral concentration that read-twice gates inherit from their layered, bounded-history dynamics, while IP_2's uniform spectrum forces exponential blow-up.

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [2105.00244v1] l1-Norm Minimization with Regula Falsi Type Root Finding Methods - [0904.1710v2] Comparison of matrix norms on bipartite spaces - [2603.18511v2] Norm-trace and Kloosterman sums in finite semi-simple algebras - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1511.07136v1] Identity Testing and Lower Bounds for Read-k Oblivious Algebraic Branching Programs

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4dc42f6c.py", line 117, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4dc42f6c.py", line 82, in run_trial
    truth_table = [simulate_bp(bp, input_) for input_ in range(1 << n)]
                   ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4dc42f6c.py", line 29, in simulate_bp
    return bp['accept_states'].get(current_state, 0)
           ^^^^^^^^^^^^^^^^^^^^^^^
AttributeError: 'set' object has no attribute 'get'
```

Judge reasoning

The test crashed with an AttributeError ('set' object has no attribute 'get') in simulate_bp before any trial completed, so no data on $\|\hat{f}\|_1$ vs (s+1)^3 was produced. The pre-registered support criterion cannot be evaluated. | next: Fix the BP data structure bug—represent accept_states as a dict (or use membership testing on a set) in simulate_bp—then rerun the full 450-seed sweep across (n,w)∈{6,8,10,12,14}×{2,3,4}.

Apolar Catalecticant Defect Bounds DISJ Randomized Communication

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Apolarity / catalecticant rank in classical secant-variety algebraic geometry (Iarrobino-Kanev; Landsberg; rarely applied to communication complexity, <5 papers) × Two-party randomized communication complexity of DISJ_n and related sign-matrix functions (Razborov 1992; Sherstov pattern matrix)
- **Recorded**: 2026-05-21 12:01 UTC
- **Entry ID**: `11bf61185540`

Statement

For any 2-party Boolean function f: {0,1}^n x {0,1}^n -> {0,1}, define the signed weight-profile 3-tensor S_f in R^{(n+1) x (n+1) x (n+1)} by S_f[i,j,k] := (1/Z_{i,j,k}) * sum_{|a|=i, |b|=j, |a cap b|=k} (-1)^{f(a,b)}, where Z_{i,j,k} = binom(n,i)*binom(i,k)*binom(n-i,j-k) is the orbit size (so S_f is a sign-mean per (i,j,k) cell, with S_f=0 if Z=0). Let kappa(f) := rank over Q of the mode-1 unfolding of S_f after thresholding singular values at 1/(n+1)^2. Conjecture (C): for every f, R_{1/3}^{cc}(f) >= (1/8) * kappa(f) * log(n) - O(log n); in particular kappa(DISJ_n) = Theta(n) (saturating the (n+1) bound) while kappa(EQ_n), kappa(GT_n with Alice<Bob trivialized), and kappa of any f with R^{cc}(f) = O(log n) are O(log n). One counterexample = one function with measurable R^{cc} <= c log(n) but kappa >= 2c log(n) + 4 falsifies (C).

Rationale

The S_n x S_n-orbit-averaged tensor S_f is the natural symmetric-group projector seen by any communication protocol whose acceptance probability is exchangeable in the indices (which all known DISJ protocols essentially are, by Razborov's symmetrization). Apolarity says the unfolding rank of S_f counts the apolar Hilbert function of the polarized form sum f(a,b) x^a y^b, an invariant of the underlying secant geometry on the Segre P^n x P^n. The conjecture predicts that this apolar defect is exactly what forces high randomized cost on DISJ while collapsing on permutation-invariant low-communication functions like EQ.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6afff1a0.py", line 95, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6afff1a0.py", line 74, in run_trial
    kappa_values.extend([kappa(f, n) for f in functions])
                         ^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6afff1a0.py", line 45, in kappa
    S = [[S_f(f, n, i, j, k) for k in range(n + 1)] for j in range(n + 1)]
                    ^

UnboundLocalError: cannot access local variable 'i' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError in the kappa function before producing any data, so neither the support nor falsification criteria can be evaluated. | next: Fix the scoping bug in kappa/S_f (pass i explicitly or correct the variable binding) and rerun the pre-registered sweep across n∈{6,10,16,24,40} with 30 seeds.

A_5 Barrington-Hash Equidistribution Gap for DISJ

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Barrington's A_5 word-encoding from NC^1=BP_5 (combinatorial group theory of word hashes into the alternating group A_5), viewed as a non-commutative discrepancy generator — rarely linked to communication complexity (<5 papers). × Two-party randomized communication complexity of DISJ_n and the rectangular structure of its 2^n × 2^n communication matrix.
- **Recorded**: 2026-05-21 12:07 UTC
- **Entry ID**: `7f1da1db4caa`

Statement

Fix Barrington generators α=(1 2 3 4 5), β=(1 3 5 4 2) ∈ A_5 whose commutator [α,β] is a non-trivial 5-cycle, and define φ:{0,1}^n→A_5 by φ(x)=γ_{x_1}γ_{x_2}⋯γ_{x_n} with γ_0=α, γ_1=β. For any Boolean matrix M:{0,1}$^{n\times\{0,1\}}$n→{0,1} with S=|supp(M)|, let ψ_M(g)=#{(x,y):M(x,y)=1, φ(x)·φ(y)=g} and D_5(M)=max_{g∈A_5} |ψ_M(g)−S/60|/S. Conjecture: D_5(DISJ_n) ≤ C·2^{−n/6} for an absolute constant C (exponential A_5-equidistribution of the disjoint-pair support), whereas every f with deterministic communication D^{cc}(f) ≤ n/10 satisfies D_5(f) ≥ 1/100 — so a single n ≤ 12 counterexample on either side refutes the bound.

Rationale

Barrington's encoding turns each (x,y) pair into an A_5 product whose distribution across the 60 group elements depends sensitively on whether the support of M decomposes as a small union of combinatorial rectangles; rectangle-structured supports concentrate on a coset of the commutator subgroup, while DISJ's support (a maximal antichain in the inclusion poset) lacks such alignment and should equidistribute. The bridge sidesteps algebrization because A_5 is non-abelian and not a polynomial extension of the Boolean ring, and it differs from prior non-abelian-Fourier attempts by working in the L^∞ discrepancy metric on group elements rather than irrep coefficients.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by RELATIVIZATION barrier